Key Stage 3
Mathematics

For Examiner's Use	
2 - 3	
4 - 5	
6 - 7	
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10 - 11	
11 - 12	
13 - 14	
15 - 16	
16 - 17	
18	
TOTAL	

Baseline Test

1 hour

Materials

For this paper you must have:

- mathematical instruments

You may **not** use a calculator

Instructions

- Use black ink or black ball-point pen. Draw
- Fill in the boxes at the bottom of this page.
- Answer **all** questions.
- You must answer the questions in the space provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work that you do not want to be marked.
- In all calculations, show clearly how you work out your answer.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer booklet.

Please write clearly, in block capitals.

Centre number ☐☐☐☐☐ Candidate number ☐☐☐☐

Surname ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐

Forename(s) ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐

Candidate signature _____

Answer **all** questions in the spaces provided.

1 Work out 28 + 57

[1 mark]

Answer _____

2 Write each list of numbers in order of size.
Start with the smallest.

2 (a)

20 005	2500	2050	25 000
_____	_____	_____	_____

[1 mark]

2 (b)

0.5	1.4	1	0.4
_____	_____	_____	_____

[1 mark]

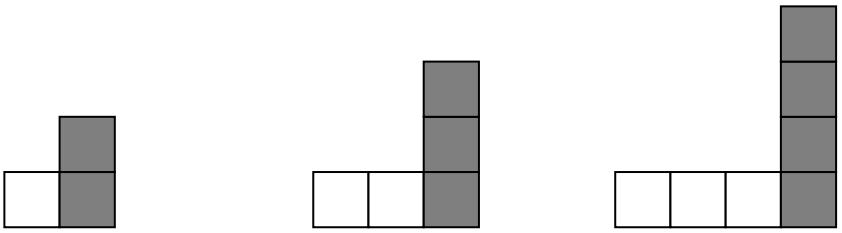
2 (c)

$\frac{5}{2}$	1	$\frac{9}{10}$	$\frac{1}{2}$
_____	_____	_____	_____

[1 mark]

3 Here are the first three patterns in a sequence.

Each pattern is made using white squares and grey squares.



Pattern 1 Pattern 2 Pattern 3

3 (a) Draw pattern 4

[1 mark]

3 (b) Work out the number of white squares and grey squares in Pattern 25

[2 marks]

White squares _____

Grey squares _____

3 (c) Priya says,

"The **total** number of squares will always be an odd number"

Give a reason why she is correct.

[1 mark]

_____

_____

_____

Turn over ▸

4 (a) Carly is thinking of a number.

Carly

When I double my number, the answer is 100

What number is she thinking of?

[1 mark]

Answer _____

4 (b) Kane is thinking of a number.

Kane

My number is between 40 and 50
It is a square number.

What number is he thinking of?

[2 marks]

Answer _____

5 (a) Here is part of the morning timetable at a school.

History	9.05 to 9.50
Maths	9.55 to 10.45

Which lesson is longer?
Show how you decide.

[2 marks]

5 (b) Afternoon lessons start at 1.50
School finishes $1\frac{1}{4}$ hours later.

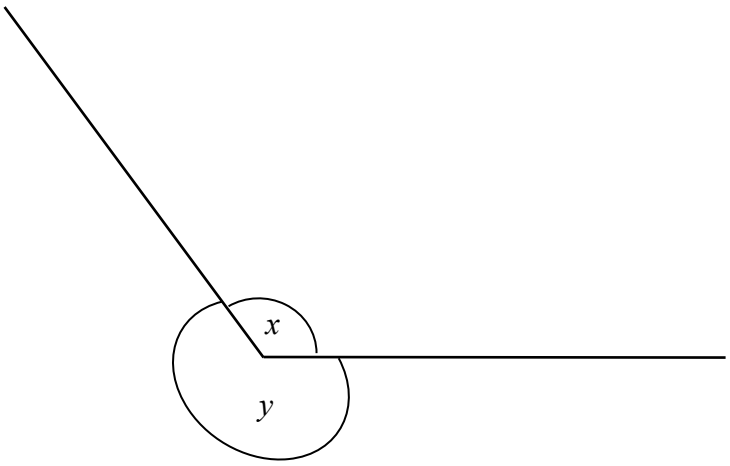
Work out the time that school finishes.

[2 marks]

Answer _____ pm

Turn over for the next question

6 The diagram shows angle x and angle y.



6 (a) Measure angle x.

[1 mark]

Answer _____ degrees

6 (b) Use your answer to (**a**) to work out the size of angle y.
You **must** show your workings.

[2 marks]

_____

_____

_____

Answer _____ degrees

7 Work out 64×28

[3 marks]

_____

_____

_____

_____

_____

Answer _____

8 Here are four number cards.

8	**2**	**5**	**9**

8 (a) Use **all four** cards to make the smallest possible number.

[1 mark]

8 (b) Use **all four** cards to make the decimal number that is closest to 60

[1 mark]

		●	

8 (c) Use **all four** cards to make the calculation true.

[1 mark]

(☐ + ☐) × (☐ – ☐) = **40**

Turn over ▸

9 (a) Work out 32×100

[1 mark]

Answer _____

9 (b) Work out $270 \div 10$

[1 mark]

Answer _____

9 (c) Work out 0.8×6

[1 mark]

Answer _____

9 (d) Work out 0.4×0.2

[1 mark]

Answer _____

10

Children need to use a child car seat if they are
between 3 and 12 years old
and
shorter than 135 cm

Do these children need to use a child seat?

Tick the correct box.

[3 marks]

		YES	NO
LIAM	**Age 10** **Height 115 cm**	☐	☐
SOPHIE	**Age 9** **Height 1 metre 38 centimetres**	☐	☐
TOM	**Age 7** **Height 0.9 metres**	☐	☐
MO	**Age 13** **Height 1.2 metres**	☐	☐
ELLIE	**Age 11** **Height $1\frac{1}{4}$ metres**	☐	☐

Turn over ▸

11 Match each circle on the left with **one** box on the right.
One has been done for you.

[3 marks]

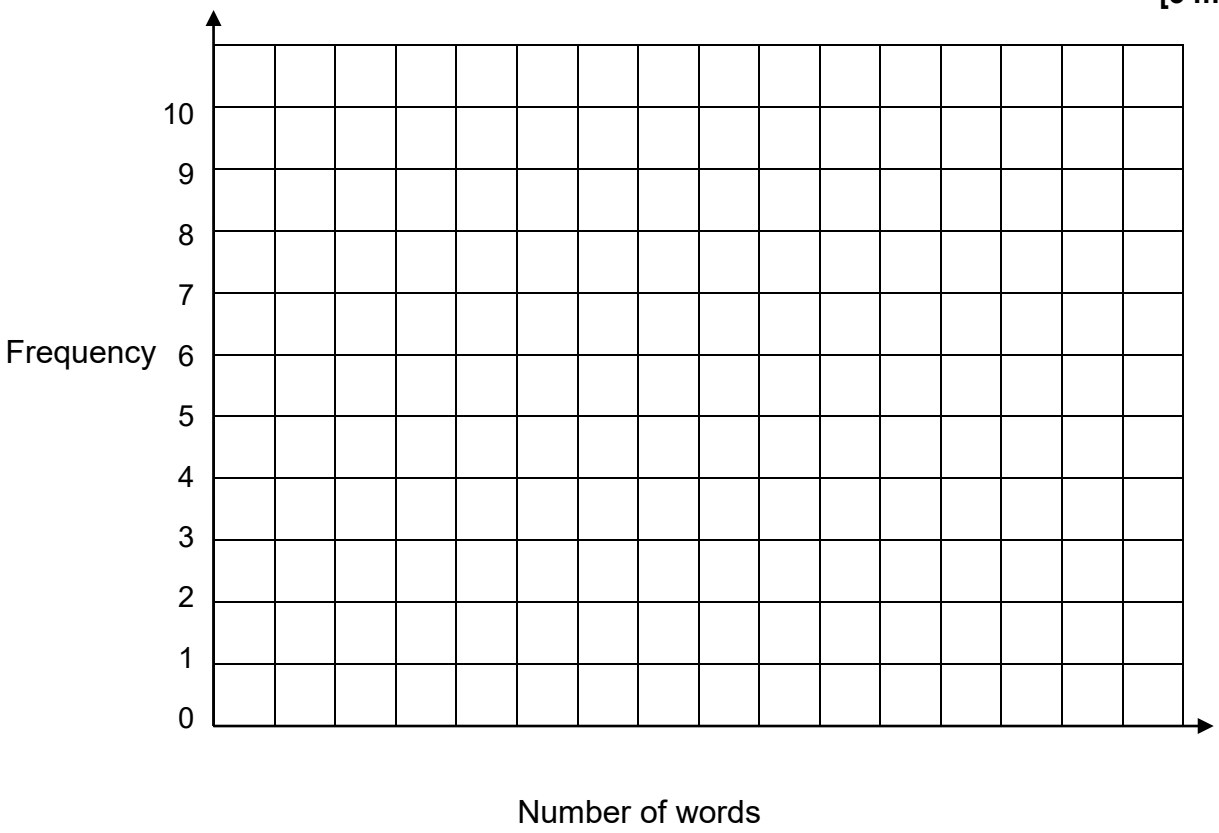
	0.25
75%	$\frac{2}{3}$
$\frac{1}{2}$ →	0.5
$\frac{4}{6}$	$\frac{3}{4}$
$\frac{3}{8}$	$\frac{12}{32}$
	0.7

12 On Monday, 24 pupils had a spelling test.
Information about the number of words they got wrong is shown in the table.

Number of words	Frequency
0	2
1	4
2	6
3	10
4	8

Draw a bar chart to show the number of words they got wrong.

[3 marks]



13 At a station,

60% of the trains arrived on time

10% of the trains arrived early.

What percentage of the trains arrived late?

[1 mark]

Answer _____ %

14 The cost of a cake is £1.30
Molly buys one cake and two cups of tea.
She pays £3.70 altogether.

Work out the cost of one cup of tea.

[3 marks]

Answer _____

15 Leroy asked his friends how many days they had been absent from school this term.
Here are the results.

1 0 4 2 0 1 1 0 0 3

15 (a) Work out the mean.
Give your answer as a decimal.

[2 marks]

Answer _____

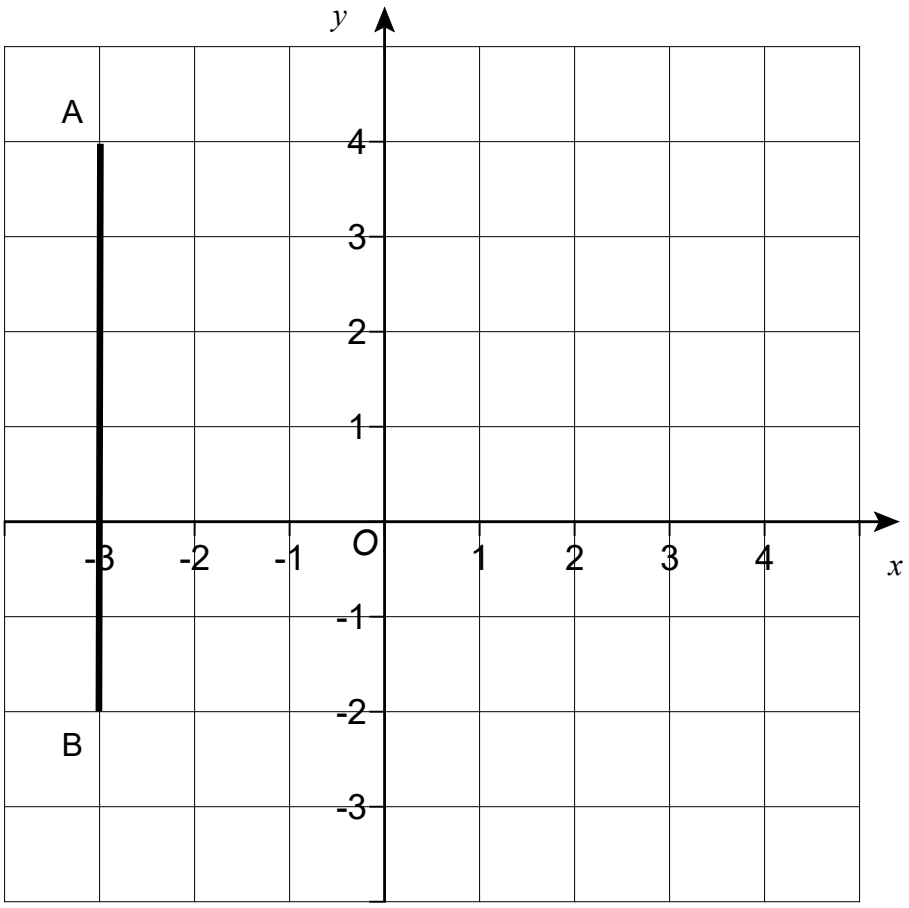
15 (b) Write down the mode.

[1 mark]

Answer _____

Turn over for the next question

16 The line from A to B is drawn on the grid.
A has coordinates (-3, 4)



16 (a) Write down the coordinates of B.

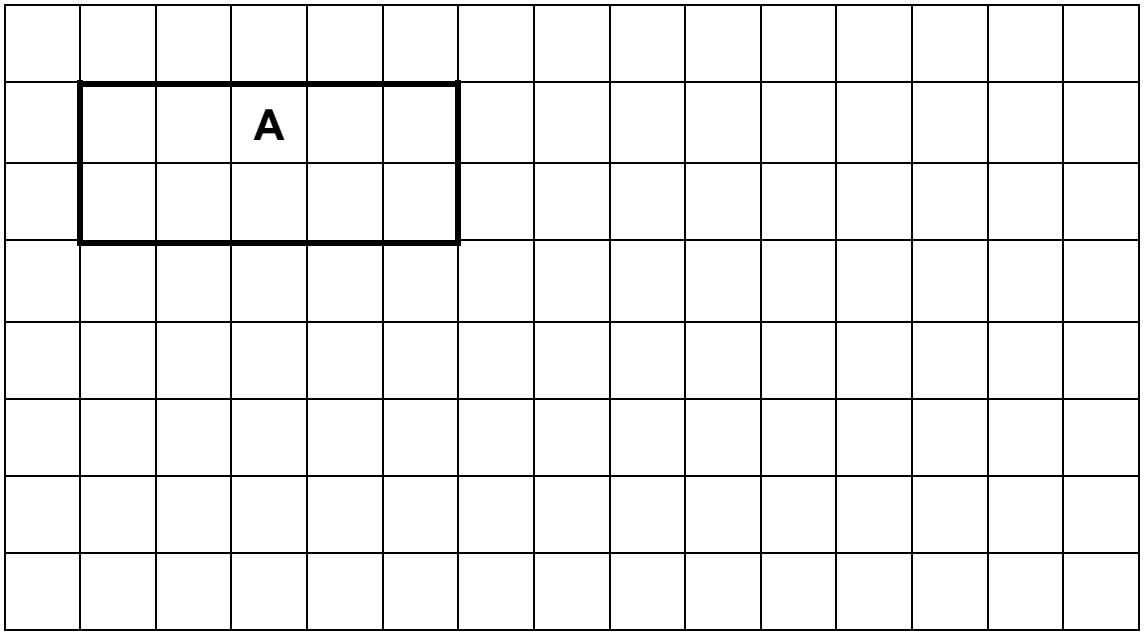
[1 mark]

Answer (_____ , _____)

16 (b) ABC is an isosceles triangle.
On the grid, mark with a cross one possible position of C.

[1 mark]

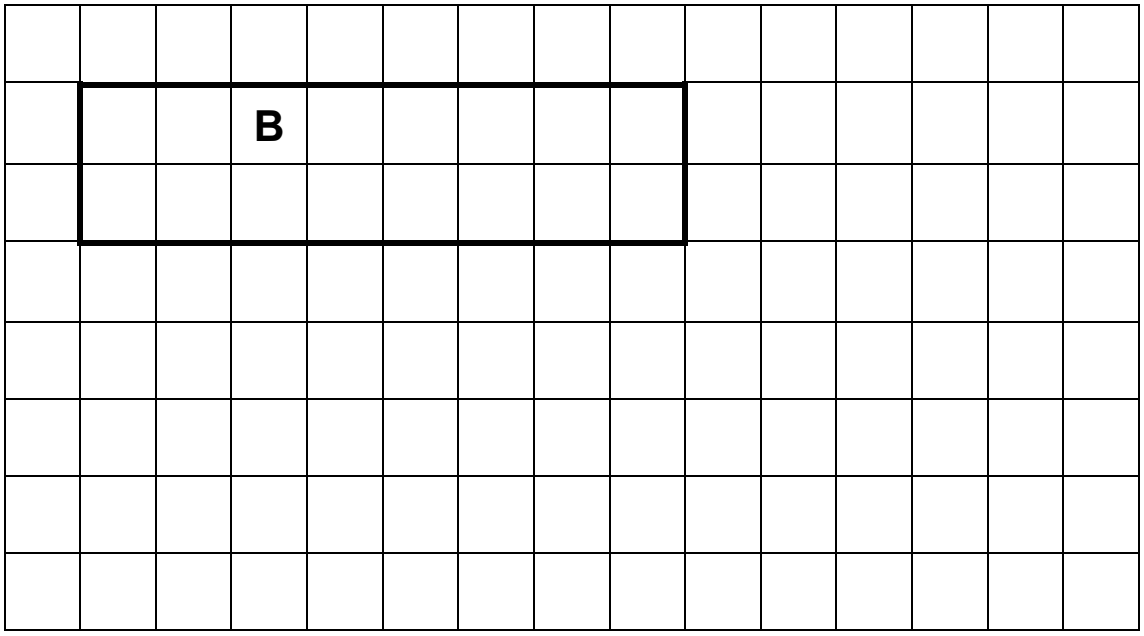
17 (a) Rectangle A has sides of length 2 cm and 5 cm



On the grid draw a **different rectangle** that has the **same perimeter** as rectangle A.

[1 mark]

17 (b) Rectangle B has sides of 2 cm and 8 cm



On the grid draw a **square** that has the **same area** as rectangle B.

[1 mark]

Turn over ▸

18 You are given that

5 miles = 8 kilometres

How many kilometres is 250 miles?

[2 marks]

Answer _____ km

19 Write one of these symbols in each box to make correct statements.

[4 marks]

= > <

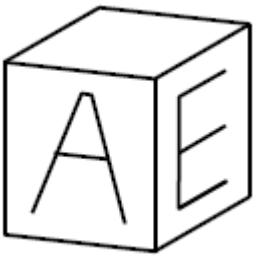
0 ☐ 3

-5 ☐ -6

-6 + 4 ☐ -2

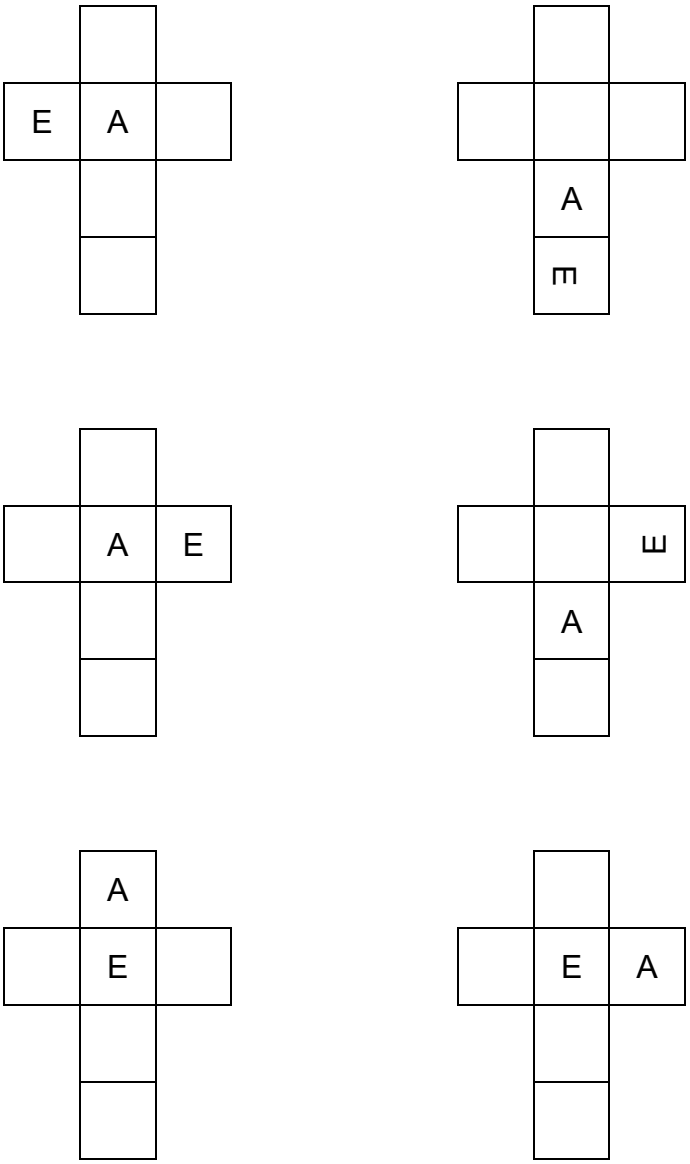
−2 – 1 ☐ 3

20 A cube has the letters A and E on two faces as shown.
The other faces are blank.



Circle the diagrams that are nets of this cube.

[2 marks]

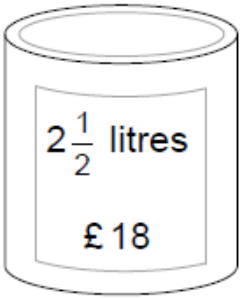


Turn over ▸

21 Rob needs 9 litres of paint.

He buys the paint in $2\frac{1}{2}$ litre tins.

Each tin of paint costs £18



Work out how much he spends.

[3 marks]

Answer £ _____

22 There are 10 red discs and 30 blue discs in a bag.
One disc is taken at random from the bag.

George says,

"There are only two colours, so the probability of taking a red disc is $\frac{1}{2}$"

Give a reason why George is not correct.

[1 mark]

END OF QUESTIONS

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DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED

Turn over

There are no questions printed on this page

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ANSWER IN THE SPACES PROVIDED